1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist framed a clear paradox — NMIBC tumors with apparently high immunogenicity (MMR/MSI features and high TIL/Immunoscore) do not always translate into durable responses to PD-1/PD-L1 blockade or other immune therapies. The key question is what tumor-intrinsic factors convert immune infiltration into functional anti-tumor immunity versus an exhausted, therapy-resistant state. 2) Tier 1 paths used: I built primarily on the MSI/dMMR → MSH6 axis (linking mismatch-repair status to neoantigen load) and the Immunoscore/PD-1–TIL augmentation paths (linking TIL density to prognosis and immunotherapy biology). These two axes directly address how tumor genomics (MMR) and immune contexture (TILs) interact to determine immunotherapy responsiveness. 3) Tier 2 support: bTMB, TMB-radiomics and liquid biopsy (CTCs) provide complementary, noninvasive readouts of neoantigen burden and clonal dynamics that could mediate or reveal the hypothesized immune-evasion (e.g., immunoediting or adaptive PD-L1 upregulation). MSH6 as an independent prognostic marker reinforces the MMR/neoantigen axis. 4) Endpoint selection: I chose therapy_response (PD-1/PD-L1 axis responsiveness) because the central puzzle concerns when favorable immune biomarkers (high TILs/MSI) actually translate to clinical benefit from checkpoint blockade or fail to do so; therapy_response is the most proximal, actionable clinical outcome for this mechanistic question. 5) Exclusions/deliberate narrowing: I excluded exploratory/weak anchors (GARS1, SARIFA-positivity, and broad 'tumor biomarkers') to keep the hypothesis sharply focused on the interactive mechanism between TIL density and tumor-intrinsic MMR/checkpoint expression. I limited core variables to one imaging feature plus two anchor genes per the Focusing Rule to produce a testable mechanistic question?

Working Hypothesis

Primary hypothesis: In NMIBC with high TIL density, tumor-intrinsic loss or dysfunction of MSH6 drives an altered neoantigen landscape that undergoes immunoediting and induces compensatory PD-L1 upregulation, culminating in T cell exhaustion and reduced clinical benefit from PD-1/PD-L1 blockade compared with MSH6-intact, similarly TIL-rich tumors. Consequently, TIL-high NMIBC tumors that are MSH6-deficient and exhibit high CD274 (PD-L1) expression will show lower therapy_response to PD-1/PD-L1 inhibitors than TIL-high, MSH6-intact tumors, thereby revealing a specific genomics–immune interaction that stratifies immunotherapy benefit. Secondary hypotheses: (1) bTMB and TMB_radiomics will correlate with MSH6 status and with PD-L1–driven exhaustion signatures, partially mediating the response disparity; (2) Immunoscore and MSI status will modulate the magnitude of the MSH6–TIL interaction; (3) Mechanistically, MSH6 loss will be associated with altered antigen presentation (e.g., HLA expression/processing genes) and an IFN-γ–driven PD-L1 axis, promoting T cell exhaustion (PD-1, LAG-3, TIM-3). The combined data will identify a patient subtype (TIL-high, MSH6-deficient, PD-L1–high) with distinct, reduced benefit from checkpoint blockade and point toward rational combination strategies to restore antigen presentation or T cell function.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSH6, CD274
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6 axis (linking mismatch-repair status to neoantigen load) and the Immunoscore/PD-1–TIL augmentation paths (linking TIL density to prognosis and immunotherapy biology).

CD274: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	bTMB	Supporting modifier from knowledge graph
Secondary	TMB_radiomics	Supporting modifier from knowledge graph
Secondary	CTCs	Supporting modifier from knowledge graph
Secondary	MSI_status	Supporting modifier from knowledge graph
Secondary	Immunoscore	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: is maintained, with secondary modifiers providing context and stratification without diluting core mechanistic inquiry.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- bTMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TMB_radiomics: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.